

Cognitive Load and Oculometrics in the Educational Scenarios: A Systematic Review and Meta-Analysis

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Cognitive Load Theory describes the concept of limited working memory capacity for information processing and cognitive tasks requiring mental effort. Cognitive load is affected by the difficulty levels involved in the learning process and mental tasks. The present meta-analysis examines the effect of difficulty levels in educational tasks based on cognitive load using fixation duration, number of fixations, and pupil dilation. The present study is based on 21 articles containing 34 comparisons between high and low cognitive load conditions in education settings. We evaluated Cohen's d for effect size, the I^2 statistic and Q-test to estimate heterogeneity, and Egger's regression test for publication bias. The heterogeneity of the experimental settings was relatively high, nevertheless, we observed meaningful differences in pupil metrics. Pupil dilation significantly increased in hard compared to easy task conditions ($d = 0.72$, $CI = 0.3672; 1.072$, p -value < 0.0001), serving as an indicator of cognitive load level in various educational settings. Fixation duration and number of fixations did not differentiate between the easy and hard levels.

CCS Concepts: • **Applied computing** → *Interactive learning environments*.

Additional Key Words and Phrases: fixation duration, number of fixation, pupil dilation, cognitive load, education

ACM Reference Format:

Merve Ekin, Krzysztof Krejtz, and Izabela Krejtz. 2025. Cognitive Load and Oculometrics in the Educational Scenarios: A Systematic Review and Meta-Analysis. *Proc. ACM Hum.-Comput. Interact.* 9, 3, Article ETRA07 (May 2025), 19 pages. <https://doi.org/10.1145/3725832>

1 Introduction

Cognitive Load Theory (CLT) defines cognitive load as the mental effort involved in performing a task [81]. Cognitive load, which limits working memory functions, can potentially influence learning outcomes. The cognitive theory of multimedia learning [60–62, 65], encompasses a range of verbal and non-verbal modes of learning material presentation, including narration, on-screen text, video, and animation [61, 65]. These dual-coding modes engage both the auditory and visual channels during learning, facilitate learning outcomes, but might negatively influence load in information processing in working memory, see also Dual-Coding Theory [15, 70].

According to CLT, lowering learning task load is possible through the modification of both, verbal and non-verbal, aspects of learning materials such as manipulating the multimedia design or

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ACM 2573-0142/2025/5-ARTETRA07

<https://doi.org/10.1145/3725832>

content of the materials [66]. Deleeuw and Mayer (2008) [18] have suggested certain strategies for reducing perceived difficulty to prevent working memory overload based on CLT, such as regulating task complexity according to the learner and/or minimizing redundant material. Consequently, a reduction in cognitive load may significantly release working memory resources required for information processing [63, 65]. Cognitive load can also be influenced by other factors, including external factors [18, 84], and the students' backgrounds [18, 84], which complicates its measurement in educational settings. However, the use of physiological measures, such as eye tracking, provides an objective metrics of cognitive load contributing to its understanding [1, 48].

Eye tracking is widely used for investigating cognitive load in different fields, including educational studies, as a non-invasive and reliable measuring method [22, 46, 103, 104]. Historically the cognitive load has been evaluated through the use of a variety of eye-movement-based metrics (oculometrics), including fixation duration, the number of fixations, and pupil size [22, 33, 46, 79]. Eye metrics have additionally been shown to differentiate cognitive fatigue [86], which shows their usefulness for predicting cognitive load levels in multimedia environments. For example, Alemdag and Cagiltay (2018) conducted a systematic review of previous studies that employed an eye-tracking method in multimedia learning context [2]. They concluded that eye movements can facilitate insight into cognitive processes during learning, including the selection, organization, and integration of information. These processes might be affected by cognitive load, which in turn influences learning performance [2].

Mutlu-Bayraktar, Cosgun, and Altan (2019) in their meta-analysis, explored the effects of task difficulty on cognitive load [66]. They concluded that task difficulty levels, such as the complexity of the material and the interactivity of the multimedia, had a significant impact on cognitive load. Another review study investigated a variety of eye measures, such as fixation duration, first fixation duration, or fixation count, for studying learning [50]. They concluded that the eye-tracking method allows for a direct connection of learning outcomes to cognitive processes. These reviews have examined the general potential of eye tracking in educational settings using qualitative evaluations. However, a focused quantitative review of cognitive load is still required. Such a quantitative summary might be achieved through the meta-analytic approach. A considerable number of studies have investigated the relationship between cognitive load and ocular metrics [1, 37, 91], there is still a lack of overall understanding of how cognitive load changes with task difficulty in educational scenarios what is an additional motivation for meta-analysis of existing research reports.

This article presents the meta-analysis of eye-tracking studies' results, discerning cognitive load levels in various educational settings. It provides insight on understanding how difficulty of educational tasks affects the learners' cognitive load and its evaluation with oculometrics.

2 Background

Up to date, several review articles demonstrated the relation between eye-movement-based metrics and cognitive load. They focused mainly on pupil dilation, fixation duration, and fixation count. A theoretical review by van der Wel and van Steenbergen (2018) assessed cognitive effort in task-evoked pupil dilation [91]. They concluded that within the three domains of cognitive control — updating, switching, and inhibition — an increase in task demands results in pupil dilation.

In their meta-analysis, Wang et al. (2024) assessed the role of pupil size, heart rate, and skin conductance in monitoring the cognitive load during driving [94]. They demonstrated quantified associations between physiological metrics, including pupil size, and different cognitive load levels in driving.

Andreou and Gkantaki (2024) investigated the use of the eye-tracking method in the reading comprehension literature in the systematic review article [3]. They demonstrated that increased cognitive effort during text processing is associated with prolonged fixation duration [3]. The

existing literature shows also that increasing task demands and expertise levels affect changes in cognitive load during a mentally demanding task. For example, Mann et al. (2007) presented the results of a meta-analysis on the cognitive performance of expert and non-expert athletes in perceptual-cognitive tasks of different sports [56]. Experts consequently had a smaller number of fixations and longer fixation durations compared to non-experts during the processing of task-related information.

It is worth noticing that some systematic reviews investigated cognitive functions via oculometrics in varying topics and experimental conditions, similar to the purpose of the present study [29, 54, 56]. Recently Bairral and Aldon (2024) in their review article examined the use of eye measurements in the context of mathematical problem-solving tasks across various educational settings [5]. Majumdar, Mondal, and Sahrawat (2022) presented a systematic review assessing oculometrics, related to fixations and pupil size, as indicators of cognitive load in various tasks and fields e.g., education, vehicle driving, product designing, human-computer interaction [54]. The results of this review showed pupil dilation to be a sensitive marker of mental effort. However, for fixation durations the review's results were mixed and inconclusive, despite the widely used *eye-mind assumption* [41] presenting theoretical basis of fixation duration and cognitive processing.

Gegenfurter, Lehtinen, and Säljö (2011) in their meta-analysis of differences between experts and non-experts in visualization comprehension in different domains (medicine, transport, and sport) demonstrated that experts, in comparison to non-experts, exhibit shorter fixation durations but higher fixations number on task-relevant areas [29]. The results are related to learning indicating that the complexity of visualizations might significantly increase a learner's working memory load.

Moreover, two systematic reviews examined the impact of various physiological measures on cognitive load, with a focus on task difficulty in educational settings [13, 89]. Tao et al. (2019) [89] conducted studies on physiological measures, including eye movements, to measure cognitive load. The results have indicated that the indicators included in the learning modeling changed following the specific task demand, task type, and level of fatigue. Charles et al. (2019) aimed to predict changes in physiological measures associated with cognitive load, with the task difficulty and task load [13]. The study comprehensively examined cognitive load differentiation via physiological measures. The authors concluded that no single measure can fully explain the comprehension of cognitive load.

2.1 Present Study

The present meta-analysis aims to contribute to a more detailed understanding of how difficulty levels influence cognitive load, measured with oculometrics, and to provide some practical recommendations for improving learner experience. To the best of our knowledge, no meta-analytic study has investigated the effect of difficulty levels on changes in cognitive load using eye-tracking metrics.

Measuring cognitive load in educational settings is complex as it is influenced by multiple factors, such as individual differences in students' prior knowledge [18, 86], external redundant elements [43, 84], and the complexity of the tasks themselves [84, 86]. Understanding fluctuations in cognitive load during tasks requiring mental effort is essential for improving instructional strategies. To achieve this, the current study uses non-invasive, psychophysiological eye-tracking metrics for assessing cognitive load levels in learning contexts. They help to overcome the subjective limitations of traditional self-report measures [85] and provide a more reliable understanding of how cognitive load changes during learning tasks with easy and hard conditions.

Our meta-analysis focuses on three oculometrics (variables): pupil dilation, fixation duration, and number of fixations. Our choice of these oculometrics was determined by the commonality of these eye-tracking measures used in the reviewed articles concerning multimedia learning domains.

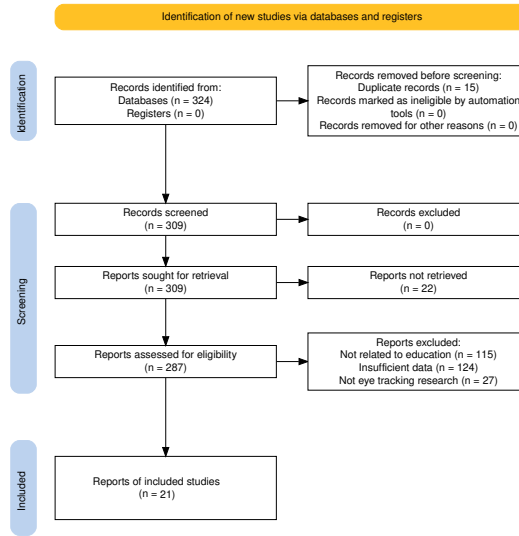


Fig. 1. Flow diagram of the articles included in the analysis based on PRISMA [31].

As well as by the requirement of at least five articles to have enough power in the random effects meta-analysis [39, 73, 90].

The oculometrics were examined in the cognitive load tasks, which were categorized according to easy and hard levels of the tasks in multimedia learning environments. The classification of the tasks as "easy" or "hard" is arbitrary and depends on the perspective of the authors of the reviewed studies. Thus, definitions of task difficulty can differ slightly across studies according to the topic and/or experimental design. The objective of comparing oculometrics in easy and hard conditions was twofold. First, we aimed to check the sensitivity and reliability of selected oculometrics in the discrimination of cognitive load levels based on the described task difficulty in the selected studies.

Second, we intended to shed light on the choice of the most promising eye-tracking measure of cognitive load for future studies. It was hypothesized that selected oculometrics would show differential patterns between task conditions "easy" and "hard", thereby allowing for the measurement of changes in cognitive load.

3 Methodology

3.1 Study Selection

PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) statement (2020) was used to guide the protocol and reporting of this systematic review and meta-analysis [69]. PubMed, Scopus, and APA PsycINFO electronic databases were searched for literature using the following formula: (("cognitive load" OR "cognitive workload" OR "mental workload") AND ("education" OR "learning") AND ("eye tracking" OR "eye movement")). This study encompasses publications published up to June 2024.

A total of 324 articles were identified and the articles found were exported to Rayyan [67], a tool for blind and collaborative article selection for systematic reviews. After removing the duplicates, 309 articles were screened. See the flowchart for the overview of the selection process (Figure 1). During the selection process, all studies were investigated by starting from the abstracts, with four criteria being given particular attention.

Studies	Selection				Comparability	Exposure			Total
	Definition of experimental and control groups	Comparability of groups	Sample Size	Volunteerism or random selection	Comparability of groups in terms of difficulty	Definition of outcomes	Measurement methods	Follow-up period	
Liu et al., 2021 [53]	*	*	-	*	*	*	*	*	7
Stolinska and Andrzejewska, 2017 [80]	*	*	*	-	*	*	*	*	7
Park and Jo, 2020 [71]	*	*	*	-	*	*	*	-	6
Tang et al., 2022 [88]	*	*	-	-	*	*	*	*	6
Zander et al., 2017 [102]	*	*	*	*	*	*	*	-	7
Dirkx et al., 2021 [21]	*	*	*	*	*	*	*	-	7
de Koning et al., 2010 [17]	*	*	*	*	*	*	*	-	7
Kuang et al., 2024 [49]	*	*	*	-	*	*	*	*	7
Ji et al., 2023 [40]	*	*	*	*	*	*	*	*	8
Que et al., 2023 [74]	*	*	*	*	*	*	*	-	7
Cao et al., 2019 [10]	*	*	*	-	*	*	*	*	7
Wu et al., 2022 [98]	*	*	*	*	*	*	*	*	8
Angraini et al., 2020 [4]	*	*	*	*	*	*	*	-	7
Bernard et al., 2021 [6]	*	*	*	*	*	*	*	-	7
Chan et al., 2022 [12]	*	*	*	*	*	*	*	-	7
Kruger et al., 2013 [48]	*	*	*	-	*	*	*	-	6
Wu et al., 2024 [99]	*	*	*	*	*	*	*	-	7
Lee et al., 2020 [52]	*	*	*	*	*	*	*	-	7
Szulewski et al., 2017 [87]	*	*	*	-	*	*	*	*	7
Castner et al., 2020 [11]	*	*	*	*	*	*	*	-	7
Lee et al., 2024 [51]	*	*	-	*	*	*	*	-	6

Table 1. Adapted Newcastle-Ottawa Quality Assessment Scale for included studies in the present meta-analysis.

First, the selected articles were published in English. Second, eye tracking was used to assess cognitive load with at least one oculometric (pupil dilation, fixation duration, or number of fixations). Third, the included studies were expected to measure cognitive load during the tasks described as easy and hard, or by expertise levels. Fourth, the experimental setups were required to be educational settings, including teaching, learning, or educational activities. If a study met all the above requirements and was available in full text, details of the results and analysis were examined. For a study to be included in the meta-analysis, it needed to present its results in a suitable manner to calculate eye-tracking metrics’ effect sizes such as means – standard deviations, *t*-test values, ANOVA (ANalysis Of VAriance) results, correlation coefficients, generalized eta square. All calculation values, used in this study, for easy and hard conditions are presented in Tables 3, 4 and 5. Some articles were excluded from the analysis based on the study’s aim for the following reasons: the analysis methods were inappropriate, there was insufficient detail in the task description about difficulty levels, or the articles could not be accessed in full-text.

The first author was responsible for conducting the literature search and data extraction, while decisions regarding inclusion were made by the authors separately. The final decision to include studies was made at a consensus meeting. Ultimately, 21 articles with 34 effect sizes were included in the meta-analysis (Figure 1). Effect sizes were calculated using a web-based effect size calculator, the practical meta-analysis effect size calculator [97].

3.2 Data Analysis and Data Quality

The present meta-analysis was performed using the *metafor* package in R, version 3.2 [64]. Owing to the heterogeneity of the oculometrics and educational tasks, the random effects model (*Restricted Maximum-Likelihood (REML) Estimator*) was selected to weight the distribution of the Cohen’s *d*

effect sizes, confidence intervals (CIs), z -values, and their significance (p -value). The weight, effect size, and confidence intervals of each study in the meta-analysis were represented in the forest plots. For an example of a forest plot, see Figure 2a. The forest plots show individual effect sizes and confidence intervals, using black squares and lines. The size of each black square represents the weight of the study in the meta-analysis. The position and width of the diamond at the bottom indicate the overall effect size and confidence interval, respectively. Forest plots are of fundamental importance in meta-analyses, as they provide a clear visual summary of the results of individual studies and their contribution to the overall effect size [9]. They help identify patterns, variability, and outlier studies.

The task conditions in the studies were evaluated based on two levels: easy and hard. The selected articles presented study conditions that differ in terms of the difficulty levels of the task. These conditions were designed to assess cognitive load based on the purpose and learning settings of the studies. In the current meta-analysis, easy and hard conditions were extracted from these studies that aimed to measure cognitive load under educational settings. Consequently, the role and definition of easy and hard conditions included in the meta-analysis were considered according to the individual criteria of each study from the cognitive load perspective. The independent variables differentiating easy-hard levels for each study are presented in Tables 3, 4, 5 in related subsections. The distinction between easy and hard conditions clarified the oculometrics associated with cognitive load differences between the task conditions in various educational scenarios, including programming, linguistic, psychology, medicine, and so forth.

For statistical analysis purposes, the experimental group was classified as a "hard" condition in the calculations, while the control group was classified as an "easy" condition. Two studies measured CL by comparing levels of expertise [11, 87]. From a meta-analytic perspective, expertise is considered a moderating variable, influencing perceived task difficulty and, therefore, cognitive load. The influence of expertise reduces the perceived difficulty of the task, which affects cognitive load in the same way as task difficulty [100]. The experienced participants were classified as the "easy" condition because of their familiarity with the tasks. Considering expertise and task difficulty as parallel domains in cognitive load models allows for more consistent comparison across studies, thus increasing the reliability and validity of meta-analyses in this area [92].

The Newcastle-Ottawa Scale (NOS) was adapted for use in this meta-analysis to evaluate the quality of the included studies [96]. The studies were evaluated according to three criteria: the selection of sample sizes, the comparability of the groups, and the definitions of exposure (see Table 1). The overall scores indicated a low risk of bias for the included studies, with a mean score of six or higher out of eight.

Statistical heterogeneity and total variability of each study were evaluated using Cochran's Q test and I^2 statistic, see Table 2. Cochran's Q test is a statistical method calculated by the differences between the observed variability in each study's result and the overall meta-analysis result to determine whether the variability between studies is greater than it would be expected by chance [16]. The I^2 statistic represents the percentage of total variation between studies that is due to true differences rather than random chance or sampling error [34, 35]. Global heterogeneity scores, as quantified by I^2 , were classified into the following categories: scores between 25% and 50% were considered as low, scores between 50% and 75% were considered as moderate, and scores between 75% and 100% were considered as high [34].

In addition, the possibility of publication bias was assessed by inspecting funnel plots and Egger's regression test. In the funnel plots, the effect size is indicated on the x -axis, with standard errors indicated on the y -axis. Each point represents one study. If the location of the points is asymmetric in the funnel, this suggests the possibility of publication bias (for an example of a funnel plot, see Figure 2b). Funnel plots are crucial in meta-analysis because they visually assess the potential

	k	Easy Condition Sample Size	Hard Condition Sample Size	Cohen's <i>d</i>	95% CIs	<i>z</i>	<i>p</i> -value	Cochran's Q	Q-test (<i>p</i> -value)	I ²	tau ²	Bias
Fixation Duration	15	363	450	1.30	−0.561; 3.151	1.37	0.17	192.67	< 0.0001	99.35%	13.19	0.003
Fixation Duration (without outliers)	12	319	403	−0.01	−0.366; 0.340	−0.07	0.94	53.87	< 0.0001	81.33%	0.30	0.72
Number of Fixations	9	263	350	0.14	−0.301; 0.580	0.62	0.53	54.96	< 0.0001	85.85%	0.38	0.62
Pupil Dilation	10	234	266	0.72	0.367; 1.072	4.00	< 0.0001	36.53	< 0.0001	70.10%	0.22	0.49

k = number of the studies, CIs = Confidence Intervals, I² = the percentage of total variation between studies that is due to true differences rather than random chance or sampling error, tau² = the variance in effect sizes between studies due to true heterogeneity, Bias = *p*-value of funnel plot

Table 2. Mean weighted effect sizes for oculometrics and cognitive load in random effects model.

for publication bias. They allow researchers to examine the symmetry of the distribution, which suggests an unbiased meta-analysis. However, asymmetry may indicate selective publication or methodological heterogeneity. They are therefore useful for assessing the validity and reliability of the results.

4 Results

4.1 Pupil Dilation in Easy and Hard Conditions of Cognitive Load Tasks

Ten studies were included in this review and meta-analysis to examine pupil dilation for easy (total sample size: *n* = 234) and hard (*n* = 266) levels of tasks to measure cognitive load in educational scenarios (see Table 3).

Studies	Educational Field	Independent Variables	Easy (n)	Hard (n)	Mean-SD (Easy)	Mean-SD (Hard)	Male (%)	Age (Mean-SD)	Eye Tracker	Sampling Rate (Hz)	Pupil Size Calculation
Liu et al., 2021	Programming	Color-coded design	19	19	3.29 ± 0.35 3.27 ± 0.31	3.17 ± 0.42 3.72 ± 0.46	61.9	20.81 ± 1.13	Tobii T120	120	Average
Que et al., 2023	Linguistic	Usage of background music	50	50	0.12 ± 0.11	0.11 ± 0.13	49.0	23.6 ± 3.9	EyeLink 1000+	N/A	Difference between average and baseline
Wu et al., 2022	Programming	Color-coded design	19	19	3.29 ± 0.35 3.37 ± 0.31	3.71 ± 0.42 3.71 ± 0.46	68.4	20.81 ± 1.13	Tobii T120	120	Difference between average and baseline
Bernard et al., 2021	Cybersecurity	Segmentation principle	10	9	2.62 ± 0.1	2.91 ± 0.09	68.4	N/A	Tobii T60	60	Average
Kruger et al., 2013	Psychology	Existence of subtitle	41	41	Mann-Whitney U (U = 178.0)	Mann-Whitney U (U = 178.0)	N/A	N/A	SMI View X	50	Difference between average and baseline
Wu et al., 2024	Learning Task	Number of turning points	15	15	ANOVA (η ² = 0.058)	ANOVA (η ² = 0.058)	33.3	28 ± 7	Tobii Pro Nano	N/A	Difference between average and baseline
Lee et al., 2020	Medicine	Existence of pause condition	30	40	0.78 ± 0.36	0.94 ± 0.28	27.1	22.7 ± 2.78	SMI RED	250	Difference between average and baseline
Szulewski et al., 2017	Medicine	Expertise level	10	13	67.73 ± 67.93	121.39 ± 113.32	65.2	36.0 ± 4.5	Tobii Glasses	30	Difference between average and baseline
Castner et al., 2020	Medicine	Expertise level	26	50	0.05 ± 0.35	0.31 ± 0.31	N/A	N/A	SMI RED	250	Difference between average and baseline
Lee et al., 2024	Medicine	Complexity of questions	14	14	0.07 ± 0.31	−0.21 ± 0.43	14.3	20.5 ± 1.5	HTC Vive Pro Eye	120	Difference between average and baseline

Independent Variables = Variables that distinguish between easy and hard conditions, **Easy (n)** = the number of easy condition sample size, **Hard (n)** = the number of hard condition sample size, **Pupil Size Calculation** = The calculation used to measure pupil dilation in the studies.

Table 3. Studies included in the meta-analysis of pupil dilation and cognitive load.

The results indicate a considerable increase in pupil dilation as an indicator of cognitive load in hard conditions compared to easy task conditions (Cohen's *d* = 0.72, 95% CI = 0.3672; 1.072, *p*-value < 0.0001) (see Table 2). Each study included in the meta-analysis of pupil dilation was presented with Cohen's *d*, their CIs, and overall effect size in the random effects model (Fig. 2a). The Cochran's Q test and the I² statistic were used to assess heterogeneity in the random effects model

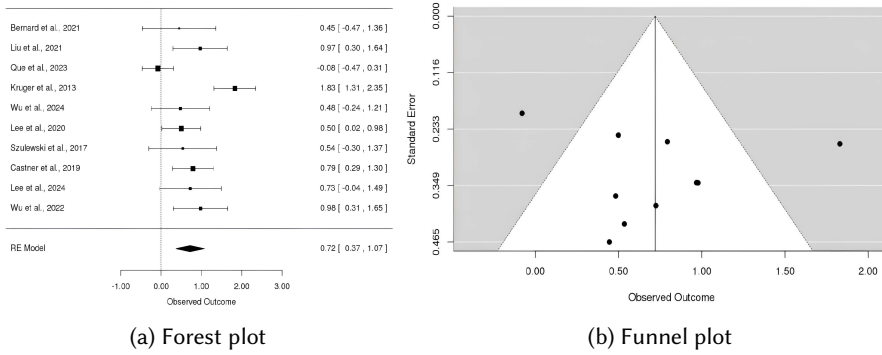


Fig. 2. Forest and funnel plots for Cohen's d effect sizes of pupil dilation in random effects model. **(2a) Forest plot.** The majority of studies showing a consistent positive effect emphasize that pupil dilation is a reliable measure of cognitive load. **(2b) Funnel plot.** The funnel plot shows that there is no publication bias, even though a slight asymmetry.

(Table 2). The results showed that the distribution of effect sizes between studies was significant ($Q = 36.53$, $df = 9$, p -value < 0.0001), and the heterogeneity value was moderate (70.10%). Therefore, it is reasonable to conclude that the random effects model is the most appropriate for analyzing the variability of effect sizes.

The risk of publication bias for studies on pupil dilation was evaluated using funnel plots and Egger's test (Fig. 2b). Egger's regression test for funnel plot asymmetry reported no publication bias ($t(8) = 0.72$, p -value = 0.49). Despite a slight asymmetry in the plot, the majority of the included studies were within the 95% CI lines, indicating that there is no risk of publication bias.

4.2 Fixation Duration in Easy and Hard Conditions of Cognitive Load Tasks

Fifteen studies were included in the meta-analysis, which was conducted to assess fixation duration in easy ($n = 363$) and hard ($n = 450$) conditions for cognitive load during educational scenarios (see Table 4).

The random effects model indicated that the difference was not significant between cognitive load levels for fixation duration (Cohen's $d = 1.30$, 95% CI = $-0.5613; 3.1531$, p -value = 0.17, see Table 2). However, we excluded three extreme effect sizes (outliers) for fixation duration and cognitive load (Fig. 3a). Outlier studies [4, 6, 12], were identified through the non-overlapping of the study confidence interval with the pooled effect confidence interval [93].

The forest plot presents Cohen's d and their 95% CI for each study analyzing fixation duration (Fig 3b). The studies showed varying effect sizes, with some reporting negative effect sizes and others reporting positive effect sizes, suggesting potential heterogeneity in the results. The heterogeneity of the distribution of effect sizes across the studies was significant ($Q = 192.67$, $df = 14$, p -value < 0.0001 , see Table 2). Furthermore, the I^2 statistic indicated that 99.35% of the total variation between studies was due to the considerable heterogeneity of studies included in the meta-analysis for fixation duration as an indicator of cognitive load.

The funnel plot assessing publication bias in the meta-analysis of fixation duration illustrates the standard error associated with Cohen's d effect sizes (Fig. 4a). The Egger's regression test for the funnel plot asymmetry was conducted to assess the presence of publication bias ($t = 3.73$, $df = 13$, p -value = 0.003). As the p -value of the regression test is lower than 0.05, this provides evidence

Studies	Educational Field	Independent Variables	Easy (n)	Hard (n)	Mean ± 1SD (Easy)	Mean ± 1SD (Hard)	Male (%)	Age (Mean ± 1SD)	Eye Tracker	Sampling Rate (Hz)
Liu et al., 2021	Programming	Color-coded design	19	19	218.12 ± 103.23	314.00 ± 137.50	61.9	20.81 ± 1.13	Tobii T120	120
Stolinska and Andrzejewska, 2017	Programming	Color-coded design and element count	23	22	254.4 ± 58.6	248.6 ± 34.3	60.0	16	SMI iView X	250
					307.5 ± 58.1	221.5 ± 28.8				
					245.6 ± 37.0	263.8 ± 47.6				
						250.6 ± 27.1				
Park and Jo, 2020	Online Course	Self-directed learning	49	60	Correlation coefficient (r = -0.204)	Correlation coefficient (r = -0.204)	N/A	N/A	N/A	N/A
Tang et al., 2022	Course Management	Presentation format	20	20	t-test (t = -4.69)	t-test (t = -4.69)	77.3	24.8	EyeLink 1000+	500
Zander et al., 2017	Human Anatomy	Personalized language	18	19	168.64 ± 22.64 80.87 ± 13.50	187.22 ± 16.54 66.32 ± 16.88	51.4	25.01 ± 3.92	EyeLink II	250
Dirkx et al., 2021	Math	Adapted learning material	29	29	353.62 ± 79.18	358.26 ± 91.51	55.2	18.77 ± 3.05	SMI RED	250
de Koning et al., 2010	Biology	Cueing of material	13	13	224.85 ± 78.05	229.12 ± 61.45	50.0	21.43 ± 2.27	Tobii 2150	N/A
Kuang et al., 2024	Psychology	Position of the instructor	17	18	454.78 ± 67.60	411.20 ± 77.51	22.6	20.43 ± 2.11	Tobii T120	120
Ji et al., 2023	Linguistic	Linguistic level of the word	31	31	180.66 ± 79.67	204.86 ± 79.69	16.1	24.33	Tobii Pro Spectrum	2000
Que et al., 2023	Linguistic	Usage of background music	50	50	83.77 ± 38.00	101.35 ± 42.14	49.0	23.6 ± 3.9	EyeLink 1000+	N/A
Cao et al., 2019	Online Course	Presence style of the instructor	31	31	141.59 ± 58.00	170.60 ± 42.43	40.3	N/A	GazeTech	120
Wu et al., 2022	Programming	Color-code design	19	19	218.12 ± 103.23	314.00 ± 137.50	68.4	20.81 ± 1.13	Tobii T120	120
Anggraini et al., 2020	Human Anatomy	Presentation style of the lecture	10	10	614.19 ± 86.46	471.38 ± 117.3	N/A	N/A	Tobii T60	60
Bernard et al., 2021	Cybersecurity	Segmentation principle	10	9	214.19 ± 20.37	299.34 ± 15.74	68.4	N/A	Tobii T60	60
Chan et al., 2022	Linguistic	Existence of subtitles	24	28	214 ± 3.62	275 ± 4.08	18.6	N/A	SMI RED	250

Independent Variables = Variables that distinguish between easy and hard conditions, **Easy (n)** = the number of easy condition sample size, **Hard (n)** = the number of hard condition sample size.

Table 4. Studies included in the meta-analysis of fixation duration and cognitive load.

of possible publication bias. This finding suggests that the asymmetry observed in the funnel plot might be due to publication bias of included articles.

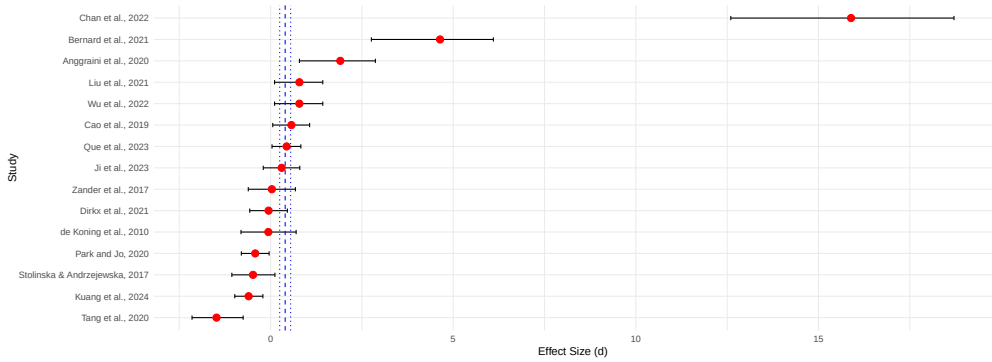
After the exclusion of outlier studies, the funnel plot exhibiting symmetry and the Egger’s test ($t = 0.37$, $df = 10$, p -value= 0.72), indicating that potential publication bias was no longer a concern (Fig. 4b). However, the model still demonstrated a non-significant and heterogeneous effect on fixation duration across difficulty levels of cognitive load (Cohen’s $d = -0.01$, 95% CI = $-0.3005; 0.5800$, p -value = 0.53, $Q = 54.96$, $I^2 = 85.85$, see Table 2).

4.3 The Number of Fixations in Easy and Hard Conditions of Cognitive Load Tasks

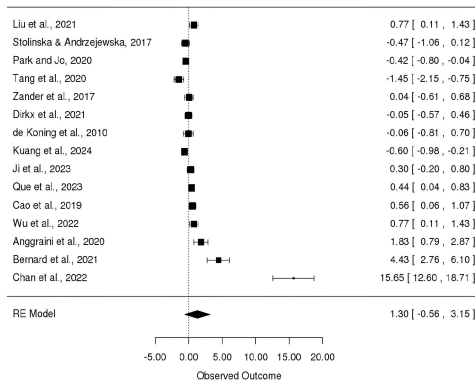
The objective of this meta-analysis was to evaluate the difference in the number of fixations for easy ($n = 263$) and hard ($n = 350$) tasks levels to understand cognitive load in educational scenarios for which effect sizes from nine studies were analysed (see Table 5). The Cohen’s d , the 95% CI for each study, and the overall effect size in the random effects model are presented in Fig. 5a.

The random effects model indicated a non-significant effect for the number of fixations (Cohen’s $d = 0.14$, 95% CI = $-0.3005; 0.5800$, p -value = 0.62, see Table 2). Cochran’s Q test showing the heterogeneity of independent studies in the random effects model, revealed significant variability in effect sizes ($Q = 54.96$, $df = 8$, p -value < 0.0001). The I^2 statistic (85.85%) also demonstrates a high heterogeneity value. The results indicated that variability in the effect sizes of the included studies supported the use of the random effects model for the meta-analysis.

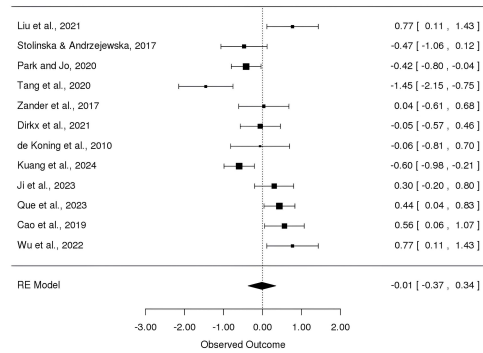
A funnel plot for the number of fixations was employed to evaluate the potential publication bias in the studies included in the meta-analysis (Fig. 5b). The asymmetry of the funnel plot was also assessed using Egger’s regression test, and the results indicated that the p -value was not significant, indicating no evidence of publication bias ($t = -0.63$, $df = 10$, p -value= 0.54). The symmetry around



(a) Screening for outlier studies with effect sizes and CIs, for fixation duration (blue dashes: the pooled CI, blue dots: the upper and lower pooled CI)



(b) Forest plot before excluding outliers



(c) Forest plot after excluding outliers

Fig. 3. Outlier analysis and forest plots for Cohen's d effect sizes of fixation duration in random effects model. **(3a) Outlier analysis.** The first three articles report extreme effect sizes compared to the pooled confidence intervals. **(3b) Forest plot before excluding outliers.** The large diamond represents the overall effect size and the wide confidence interval reflects significant variability due to the outlier studies. **(3c) Forest plot after excluding outliers.** The smaller diamond and narrower confidence interval indicate reduced variability and a nearly zero effect size, indicating a pattern after outliers are removed.

the vertical line (overall effect size) in the funnel plot provides evidence that publication bias is not a factor affecting the results.

5 Discussion, Limitations, and Conclusions

This review and meta-analytical article analyzed 34 Cohen's d values derived from 21 studies focusing on the distinctions in oculometrics (pupil dilation, fixation duration, and fixation count) related to the described difficulty of mental tasks within educational environments, as indicators of cognitive load. To the best of our knowledge, this is the first meta-analysis to address this purpose.

This meta-analysis included 10 Cohen's d for pupil dilation, easy ($n = 234$) and hard condition ($n = 266$), 15 Cohen's d for fixation duration, derived from easy ($n = 363$) and hard condition ($n = 450$), and nine Cohen's d for the number of fixations, easy ($n = 263$) and hard condition ($n = 350$).

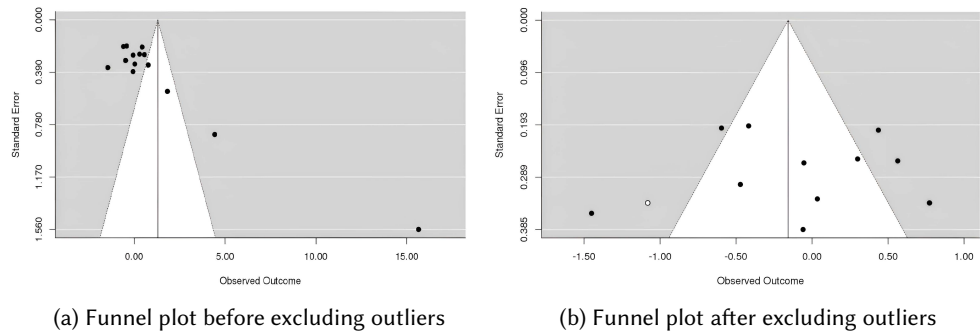


Fig. 4. Funnel plots for Cohen’s *d* effect sizes of fixation duration in random effects model. **(4a) Funnel plot before excluding outliers.** The extreme points far to the right of the funnel indicate outliers that cause asymmetry and heterogeneity in the data. **(4b) Funnel plot after excluding outliers.** The points are more equally distributed around the center of the overall effect size line, creating a more symmetrical funnel shape, suggesting improved homogeneity and less bias.

Studies	Educational Field	Independent Variables	Easy (n)	Hard (n)	Mean ± 1SD (Easy)	Mean ± 1SD (Hard)	Male (%)	Age (Mean ± 1SD)	Eye Tracker	Sampling Rate (Hz)
Stolinska and Andrzejewska, 2017	Programming	Color-coded design and element count	23	22	45.9 ± 30.0 120.2 ± 84.7 141.7 ± 85.7	35.9 ± 10.5 94.7 ± 87.3 122.0 ± 84.1 157.4 ± 89.0	60.0	16	SMI iView X	250
Park and Jo, 2020	Online Course	Self-directed learning	49	60	Correlation coefficient (r = -0.321)	Correlation coefficient (r = -0.321)	N/A	N/A	N/A	N/A
Tang et al., 2022	Course Management	Presentation format	20	20	t-test (t = -3.89)	t-test (t = -3.89)	77.3	24.8	EyeLink 1000+	500
Zander et al., 2017	Human Anatomy	Personalized language	18	19	76.88 ± 2.72 30.95 ± 4.07	81.19 ± 3.40 24.70 ± 5.51	51.4	25.01 ± 3.92	EyeLink II	250
Kuang et al., 2024	Psychology	Position of the instructor	17	18	453.65 ± 220.34	633.5 ± 263.68	22.6	20.43 ± 2.11	Tobii T120	120
Ji et al., 2023	Linguistic	Linguistic level of the word	31	31	2.02 ± 1.32	2.64 ± 1.44	16.1	24.33	Tobii Pro Spectrum	2000
Que et al., 2023	Linguistic	Usage of background music	50	50	381 ± 162	457 ± 185	49.0	23.6 ± 3.9	EyeLink 1000+	N/A
Cao et al., 2019	Online Course	Presence style of the instructor	31	31	331.23 ± 91.97	419.35 ± 96.09	40.3	N/A	GazeTech	120
Chan et al., 2022	Linguistic	Existence of subtitles	24	28	3.54 ± 3.34	5.29 ± 3.66	18.6	N/A	SMI RED	250

Independent Variables = Variables that distinguish between easy and hard conditions, **Easy (n)** = the number of easy-condition sample size, **Hard (n)** = the number of hard-condition sample size.

Table 5. Studies included in the meta-analysis of the number of fixations and cognitive load.

The analyses for the random effects models revealed that pupil dilation was significantly higher in the hard conditions of tasks, requiring more mental effort than in the easy conditions ($d=0.72$). Further, we demonstrate that the effect sizes for fixation duration ($d=1.30$) and the number of fixations ($d=0.14$) were not significant. The effect for fixation duration remained insignificant, even after three outlying studies were removed ($d=-0.01$). These findings suggest that out of the three oculometrics considered in the present meta-analysis, pupil dilation might serve as a reliable and sensitive indicator of changes in cognitive load related to task demands in educational settings.

The meta-analysis was based on studies with various designs. That reflects the diversity of contexts in educational and social empirical investigations [38] and they all contribute to the understanding of cognitive load theory. It also gives a broader perspective on the applications of Cognitive Load Theory [82] in education practice.

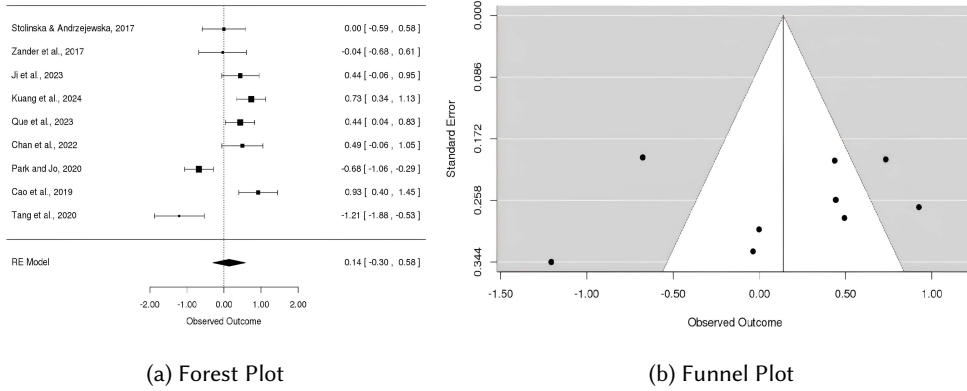


Fig. 5. Forest plot and funnel plot for Cohen's d effect sizes for fixation number in random effects model. **(5a) Forest plot.** The diamond at the bottom shows the overall effect size, and most studies also cluster around the null. This indicates a minimal significant effect of cognitive load on the number of fixations across studies. **(5b) Funnel plot.** The funnel plot shows a symmetrical pattern with points equally distributed around the center, indicating that there is no publication bias.

Our purpose was to bring together these different perspectives on cognitive load, and to provide a novel insight into the robustness of oculometrics in differentiating between cognitive load levels. However, we are aware that the heterogeneity of included studies might influence the insignificant results for fixation duration and fixation count, and further investigations should be conducted for dissolving the nuances in which these measures might shed light on differences in task difficulty.

Pupil dilation is widely used for measuring cognitive load starting from a classical article by Hess & Polt [33] (for a review, see der Wal & Steenbergen [91]). The present meta-analysis demonstrated a statistically significant effect size of 0.72 for pupil dilation, indicating an increase in pupil size in hard conditions. This measure is the only one of the three selected oculometric measures to show significance in this meta-analysis.

Previous research has indicated that different methods of measuring pupil size, such as the Index of Pupillary Activity (IPA) [23], the Low/High Index of Pupillary Activity (LHIPA) [24], and task-evoked pupillary responses [59] may also offer insights into cognitive processes. These measures can be employed to assess cognitive load also in real-time.

We should, however, also notice some limitations of pupillary activity measures. Pupil dilation is altered by ambient light changes [42]. For example, Pfleging et al. (2016) [72] conducted a study measuring cognitive load under controlled lighting conditions for designing a predictive pupil diameter model. They achieved 75% accuracy in predicting cognitive load with pupil dilation in their model, even when different lighting conditions were taken into account. Their results also highlighted the effect of light on pupil size, which changed independently of cognitive load. To control for the lighting, researchers employ baseline correction techniques, which involve the subtraction of the task condition from the baseline condition [57]. Eight out of ten studies in the present meta-analysis calculated the difference between average pupil size and baseline. The pupil measurement is also affected by off-axis distortion while measuring with remote eye-tracking [46, 58], which leads to the incorrect manifestation of pupil diameter. In addition to external and measurement-related factors, pupil size can also be affected by emotional arousal [8, 47].

The intriguing result of the present meta-analysis regards fixation duration [76]. The initial results of the present meta-analysis (before removing outlying studies) showed an insignificant effect size of 1.30 for fixation duration. This may suggest longer fixation durations in more challenging educational tasks compared to easier tasks. However, after excluding outlying studies [4, 6, 12], the effect size shifted to a negative value, yet still insignificant. Despite the finding of no effect for fixation duration in the present study, another meta-analysis revealed significant expertise-related differences in the sports domain [56].

Although longer fixation durations are typically associated with increased visual information processing following *eye-mind assumption* [41], previous findings on the relationship between fixation duration and cognitive load have yielded mixed results. Some studies indicate a decrease in fixation duration with task demand, psychological stress, and difficulty [27, 55, 75] but some other studies demonstrate increase in fixation duration [20, 44, 45]. The relationship between fixation duration and cognitive load might depend also on how *easy* and *hard* tasks are defined. For example, in visual search tasks shorter fixation durations were observed during more complex tasks [26, 36]. Furthermore, Tao et al. (2019) [89] reviewed studies that reported a reduction in fixation duration with increasing difficulty levels. The other reasons for inconsistencies in the literature might come from the definitions of the cognitive load and learning. Assuming *the eye-mind relation* [41], an increase in cognitive processing of visual information does not necessarily tend to increase in cognitive load. The increase in cognitive load should result in the depletion of cognitive resources and the ability to process currently attended information. However, an increase in information processing might as well be an effect of increased engagement leading to better learning. For example, the *desired difficulty* concept in educational settings [7] tends to suggest that reaching a certain threshold of difficulty varied across individuals, is necessary for effective learning. Some easy tasks might be "difficult enough" for some participants to engage them in deep processing and increase their fixation duration during learning. Consequently, we postulate that future studies should control the desired difficulty level when implementing learning tasks.

Finally, the results showed a statistically insignificant effect size of 0.14 for the number of fixations. The number of fixations can be used to measure shifting in visual attention, indicating the level of engagement or cognitive load required to complete a task or information processing [14, 19]. In educational scenarios, the higher number of fixations and more saccadic movements may represent a deeper visual processing in challenging learning material to integrate the information together [19]. The high number of fixations reflects an increase in task complexity, thereby indicating that the visual system is actively scanning the task [95, 101]. Even though the present meta-analysis did not find significant differences in the number of fixations between easy and hard conditions of educational tasks, the relationship between the number of fixations and cognitive load in learning settings needs further exploration.

Task load, defined as a measure of performance based on the levels of complexity in tasks requiring mental effort, and cognitive load are closely related but distinct terms [32, 82]. The use of difficulty levels may also influence task load and performance. Cognitive load is employed to describe the general cognitive resources [68, 84], whereas task load is used to refer to the level of difficulty in performing a specific task [28]. In this meta-analysis, the included studies have certain educational tasks, including difficulty levels, or expertise levels. Depending on the educational tasks, the differences in oculometrics between easy and hard conditions may result from task load as well.

5.1 Limitation and Future Studies

A potential limitation of this review and meta-analysis is that measuring cognitive load in multi-media learning environments is challenging because of numerous factors that can influence and

interact with cognitive load in different study designs. These factors include individual differences in prior knowledge, task complexity, and multimedia designs. In this meta-analysis, difficulty levels are emphasized to understand and differentiate cognitive load, as task difficulty has a major impact on changes in cognitive load in educational settings [83]. However, assessing cognitive load based on multiple factors may provide a more comprehensive approach to assessing cognitive load in multimedia learning environments in future reviews.

There are convincing reports from empirical experiments showing a relationship between task loads and cognitive load and characteristics of fixational eye movements, e.g. microsaccades [46, 77, 78]. The present meta-analysis did not include those measures which is why we postulate to have them studied in future systematic reviews as the popularity and affordability of high-speed eye trackers that can measure micro eye movements is increasing.

Cognitive load can also be measured using several physiological indicators, including other than eye tracking metrics, galvanic skin response, electroencephalogram, and heart rate variability [1, 25, 30]. Future systematic reviews and meta-analyses may be enhanced by understanding the implications of a multi-physiological approach to the assessment of cognitive load in multimedia environments. Furthermore, the relationship between physiological measures and different types of cognitive load (intrinsic, extraneous, and germane) may provide deeper insights into educational strategies and engaging learners.

5.2 Practical Implications and Conclusions

The present findings emphasize the role of oculometrics, particularly pupil dilation, in understanding and assessing cognitive load in various educational settings. These findings have practical implications for educators and instructional designers, as eye-tracking data can provide real-time, objective feedback on students' learning processes and cognitive load levels [68]. In addition, by measuring eye tracking in a real-time environment, we may provide individualised assessments and help improve educational outcomes. By incorporating eye-tracking technology into educational scenarios, educators can more effectively design materials based on cognitive capacity, prior knowledge, and dynamic changes in cognitive load. In addition, learning could be enhanced by regulating task complexity, reducing redundant elements, and monitoring schema construction which means considering cognitive load types that are intrinsic, extraneous, and germane [82, 104].

This meta-analysis supports the growing evidence for using eye-tracking metrics as reliable indicators of cognitive load in educational settings. The results suggest that eye tracking can serve as a valuable tool for distinguishing cognitive load levels with increasing task demands. Although oculometrics were found to be able to predict task demands by evaluating cognitive load, its reliability was not consistently evident across all instructional contexts because of the nature of the tasks' differences. Future research should explore how eye metrics, especially pupillary response, can be integrated into learning systems to individually assess the dynamics of cognitive load.

6 Privacy and Ethics Statement

This meta-analysis is based on data from previously published studies. No new data were collected from human participants. We address that there are no impacts of this meta-analysis on the authors' interpretation of privacy, fairness, safety, human rights, data sovereignty, or future misuse, and any resulting risk-benefit trade-offs.

7 Acknowledgement

This work was funded by the European Union through the MSCA Doctoral Network as part of the Eyes for Interaction, Communication, and Understanding (Eyes4ICU) project (Grant No: 101072410).

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Received November 2024; revised February 2025; accepted March 2025